Prifysgol Wrecsam
Wrexham University

MODEL CARD:

MATERNAL HEALTH RISK PREDICTION (CATEGORIZATION) APP

BY

NAME: PEACE EPUEH-OGHENA ASOGBHA

Model Overview and Intended Use

Model Overview

Maternal Risk Prediction System is a decision-support Machine Learning (ML) web-based App developed to complement clinical efforts targeted at ascertaining an individual's susceptibility to maternal health risks, especially in resource-limited healthcare settings. Maternal health remains a global concern, and studies [2], [3] showed that maternal morbidity and mortality can be significantly reduce when physiological risk factors, such as abnormal blood pressure, elevated blood glucose, and irregular heart rate are identified early.

This system implements that principle by using a classification-based ML model trained to estimate a woman's level of maternal risk (Low, Medium, High) based on specified physiological indicators.

The deployed model relies on the Random Forest algorithm, which was selected due to its strong predictive performance, robustness, and interpretability under multiple evaluation metrics.

Intended Users and Deployment Context

The system is designed primarily for:

- Healthcare professionals (nurses, midwives, clinicians).
- Community health workers in low-resource settings.
- Caregivers supporting maternal health monitoring.

It is deployed as a web-based application, enabling accessibility in settings where advanced medical diagnostic infrastructure may be limited. The system is particularly suited as part of:

- Pre-clinical screening and triage.
- Routine antenatal monitoring.
- Holistic clinical decision-making context.

Appropriate Use Cases

Basically, the model is intended to:

- Complement early detection of maternal health risks.
- Support the prioritization of patients requiring additional medical care.
- Enhance awareness of physiological risk factors during pregnancy.

Interestingly, the prediction (output) is accompanied by concise patient-friendly explanation of an individual's risk level.

Non-Intended Uses

The predictions are made exclusively on the basis of input features and without taking into account users' full medical history and contextual clinical factors. Hence, this system is not intended to:

- Replace professional medical diagnosis or clinical judgment
- Serve as a standalone healthcare support tool.
- Be used in high-stakes emergency scenario without medical expert guidance.

System and Data Summary

Model Type, Input and Output Variables

The predictive capability of this ML classification model is predicated on a tree-based ensemble learning technique, precisely *Random Forest*. Apparently, the model takes structured clinical input data and gives a categorical prediction (output) as the level of maternal risk. Model's variables are delineated in Table 1.

Table 1 Characteristics of project formulation dataset

VARIABLE	ROLE	DESCRIPTION	UNITS
Input Variables			
Age	Feature	Any age in years of a woman during pregnancy.	years
Body Temperature	Feature	The body temperature of the woman during pregnancy.	°F
Blood Sugar	Feature	Blood glucose levels in terms of a molar concentration	mmol/L
Diastolic BP	Feature	Lower value of blood pressure.	mmHg
Systolic BP	Feature	Upper value of blood pressure.	mmHg
Heart Rate	Feature	A normal resting heart rate	bpm
Target Variable			
Risk Level	Target	The predicted risk intensity level during pregnancy. Where each level is explicated as follows: ▪ Low risk: Indicates that the pregnancy is considered healthy with minimal complications. ▪ Medium risk: Indicates that there might be some mild health concerns that should be monitored. ▪ High risk: Indicates significant health concerns that could lead to complications for the mother or baby; requiring close monitoring, special care, or interventions.	N/A

Data Summary and Assumptions

The model was trained on a dataset sourced from the UCI Machine Learning Repository, consisting of 1,014 observations collected from hospitals, clinics, and maternal healthcare centres in rural Bangladesh using IoT-based monitoring systems [1].

The dataset is representative of the variability in the physiological profiles of the target population (pregnant women); underscoring its practical relevance.

Key assumptions include:

- Original dataset is correct and clinically-measured.
- Each observed cases represent an independent measure.
- Similar physiological patterns correspond to similar risk profiles.

Evaluation and Performance Summary

Evaluation Approach

Model performance was assessed using a combination of standard classification metrics and validation techniques to ensure both accuracy and robustness. These included:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Receiver Operating Characteristic (ROC-AUC)
- K-Fold Cross-Validation

Each model (Random Forest, XGBoost, and CatBoost) was evaluated using identical preprocessing pipelines to ensure fair comparison.

Performance Summary

Across all evaluation metrics, the *Random Forest* model showed good and balanced results. Although, in terms of its K-Fold Cross-Validation, comparative analysis (as seen in Table 2), showed that *XGBoost* was the best performer, however, *Random Forest* was fairly stable to multiple data splits. This substantiates the model's predictive capability to unseen data, as against being too sensitive to predefined train datasets.

Evidently, *Random Forest* achieved a relatively high Accuracy score, in addition to maintaining better Precision and Recall scorers across all risk categories; highlighting its ability to accurately differentiate among the 3 risk levels.

Confusion matrix analysis showed that the model effectively minimized misclassification, especially in high-risk cases, which are of critical importance in a clinical setting. Also, its ROC-AUC output revealed that there was a high separability of the risk classes, which also supports the predictive capacity of the model.

Table 2 Comparative performance of ML algorithms across evaluation metrics

EVALUATION METRIC	SIGNIFICANCE	MODEL SCORES		
		Random Forest	***XGBoost***	***CatBoost***
Accuracy	• Measured the overall correctness of the model. • Significant in delineating the percentage of correct predictions.	**0.857143**	**0.857143**	0.852217
Precision	• Indicated how many of the predicted positive cases for each class are actually correct. • Significant in reducing false predictions.	0.857215	**0.857406**	0.853437
Recall	• Measured the model's ability to correctly identify all actual cases of each class. • Significant in identifying high-risk patients.	**0.865589**	0.864729	0.860614
F1-score	• Mediated between Precision and Recall. • Significant in balancing the trade-off between them for each risk level.	**0.860623**	0.860300	0.856528
Confusion Matrix	• Provided detailed class-level insight. • Significant in delineating true positive, false positive, true negative, and false negative cases.	**174 / 203**	**174 / 203**	173 / 203
AU-ROC (Avg Across Classes)	• Measured the class differentiation ability of the model.	**0.956**	0.950	0.947
K-Fold Mean Accuracy (Std Dev)	• Assessed the 3 models' performance on different subsets of the entire dataset. • Significant in reducing classification bias as it provided validation on different segments of the data.	0.8353 (0.0239)	**0.8481 (0.0214)**	0.8353 (0.0266)

Although the results of the evaluation are encouraging, the foregoing limits the app's flexibility and usability:

- App operates within predefined user input value ranges for each feature.
- Model evaluation does not incorporate objective clinical parameters.

Limitations, Risks and Responsible Use

Model Limitations

Given that the model is constrained by the nature and extent of the input data. It depends only on a set of predefined clinical features and does not take into consideration:

- User medical history.
- Socioeconomic factors
- Genetic predispositions
- Environmental influences

Consequently, predictions are to be perceived as indicative rather than definitive.

Potential Risks

The primary risks associated with the system include:

- Misclassification of high-risk patients as low risk.
- Over-reliance prediction results by non-expert users.
- Use outside intended clinical contexts

These could to lead to erroneous medical intervention.

Ethical and Societal Considerations

Given its application in maternal healthcare, the system should be utilized with a close regard to:

- General wellbeing of the users.
- Equitable medical care.

There is could be risk of algorithmic bias, especially if the model is applied to populations that differ ‘significantly’ from the train data conditions.

Responsible Use Guidelines

To ensure safe and effective application:

- The system should be used as a support tool, not a replacement for standard clinical practice.
- Predictions should always be interpreted alongside professional (medical) advice.
- Users should be trained to understand the system's capabilities and limitations.

The inclusion of patient-friendly comments (after each prediction) is intended to enhance understanding, while reinforcing the advisory role of the output.

Summary

The Maternal Risk Prediction System represents a practical application of machine learning in healthcare, offering a scalable and accessible tool for early maternal risk detection. Although it is shown to have good performance and usability, its responsible use must take into account its limitations and the context of the whole clinical practice where it is used.

REFERENCES

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